Generative AI in SQL and Database Testing

Here are some popular tasks and related prompts for validating databases and SQL queries.

1. Data Integrity Validation

Task: Verify that the data in the database meets predefined constraints (e.g., primary keys, foreign keys, unique constraints, and nullability).
Prompt:

- "Write a SQL query to check for duplicate records in the `users` table based on the `email` column."
- "Validate that all rows in the `orders` table have a corresponding `customer_id` in the `customers` table."
- "Check for null values in the `order_date` column of the `orders` table."

2. Query Accuracy Validation

Task: Ensure that SQL queries return the expected results.
Prompt:

- "Write a SQL query to calculate the total revenue for each customer in the `orders` table and compare it with the expected results."
- "Validate that the `COUNT` of rows in the `products` table matches the expected number of products."
- "Write a query to find all orders placed in the last 7 days and verify the output against a known dataset."

3. Performance Validation

Task: Validate the performance of SQL queries and ensure they execute within acceptable time limits.
Prompt:

- "Analyze the execution plan for a query that joins the `orders` and `customers` tables and suggest optimizations."

- "Write a query to find the top 10 most expensive queries in terms of execution time from the database logs."
- "Validate that the indexing strategy on the `orders` table improves query performance for date-range searches."

4. Schema Validation

Task: Ensure the database schema matches the expected design.
Prompt:

- "Write a SQL query to list all tables and their columns in the database and compare it with the expected schema."
- "Validate that the `orders` table has the correct data types for all columns (e.g., `order_id` is an integer, `order_date` is a datetime)."
- "Check if all required indexes exist on the `customers` table."

5. Data Consistency Validation

Task: Ensure data consistency across related tables or systems.
Prompt:

- "Write a SQL query to compare the `total_amount` in the `orders` table with the sum of `amount` in the `order_items` table for each order."
- "Validate that the `inventory` table is updated correctly after an order is placed in the `orders` table."
- "Check if the `status` column in the `orders` table matches the corresponding status in the `order_history` table."

6. Boundary Condition Validation

Task: Test edge cases and boundary conditions in the database.
Prompt:

- "Write a SQL query to find orders with a `total_amount` of 0 or negative values."
- "Validate that the `users` table handles extremely long strings in the `username` column correctly."
- "Check if the database correctly handles timezone differences in the `order_date` column."

7. Data Migration Validation

Task: Validate data after migrating from one database to another.
Prompt:

- "Write a SQL query to compare the row counts of the `customers` table in the source and target databases."
- "Validate that all data in the `orders` table was accurately migrated by comparing specific columns (e.g., `order_id`, `order_date`, `total_amount`)."
- "Check for discrepancies in data types between the source and target databases."

8. Security Validation

Task: Ensure the database is secure and adheres to access control policies.
Prompt:

- "Write a SQL query to list all users with administrative privileges in the database."
- "Validate that sensitive columns like `password` in the `users` table are encrypted."
- "Check if the database logs all failed login attempts."

9. Data Transformation Validation

Task: Validate data transformations during ETL (Extract, Transform, Load) processes.
Prompt:

- "Write a SQL query to verify that the `total_amount` in the `orders` table is correctly calculated from the `order_items` table."
- "Validate that all NULL values in the `address` column of the `customers` table were replaced with 'N/A' during the ETL process."
- "Check if the `order_date` column was correctly transformed from UTC to the local timezone."

10. Regression Testing

Task: Ensure that new changes (e.g., schema updates, query optimizations) do not break existing functionality.
Prompt:

- "Write a SQL query to compare the results of a modified query with the results of the original query."

- "Validate that the new index on the `orders` table does not affect the performance of existing queries."
- "Check if the new column `discount_code` in the `orders` table is populated correctly for all historical records."